

Which HCI Lab in South Korea Fits Best for Me? : Designing an HCI Lab Browsing System to Support Prospective Students' Decision-Making Process

Designing an HCI Lab Browsing System

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ABSTRACT

Prospective students looking for a human-computer interaction (HCI) laboratory that fits their needs and goals go through a challenging decision-making process. Often, labs and their body of work are ambiguous to the student. In this study, we aim to (1) design an HCI Lab browsing system named <HCI LAB DIRECTORY> and (2) investigate how the system supports decision making. We designed a system that can browse laboratories through research topics based on author-defined keyword (ADK) data from publications. We then conducted a user study, including an information browsing task and in-depth interviews. Findings show that our system supported decision making in (1) identifying what to look for at the initial stage and (2) developing one's own criteria by gaining saliency of the field at the drill-down stage. Our findings provide empirical understanding regarding decision-making theory and presents a new browsing system that organizes scattered and unstandardized information.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

information browsing system, information management, decision making, recommendation system, author-defined keyword analysis, HCI research lab

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1 INTRODUCTION

Looking for a human-computer interaction (HCI) graduate school laboratory that fits prospective students' needs and goals involves challenging information behavior. This is because the target is an ambiguous and undefined item. You want to find the HCI lab fits for you, but you do not know what to look for or how to search. There is neither prevalent browsing structure nor guidance from common browsing experience. Therefore, you need to go through a complex decision-making process that consists of determining saliency of the field, exploring alternatives, setting own criteria, and continuously comparing them.

The first reason that the target is ambiguous is due to the interdisciplinary nature of the field of HCI. HCI labs do not exist in a standardized form and its research topic changes over time with the advent of new technology [8]. The fact that 1,365 publications accepted in CHI Conference 2020 were assigned to 19 different subcommittees, whose topics cover from engineering technologies to design methodology, demonstrates its strong multi-disciplinarity. As such, numerous laboratories have diverse research topics and approaches. In addition, an underlying system for organizing interdisciplinary topics does not yet exist in academia.

Another reason for the difficulty is that the lab naming system is also neither standardized nor established. The name of the lab is often arbitrary. Consequently, it is difficult to guess the research topic by the name alone and decide whether to drill-down or not. Even if the naming process could be controlled, there would be intrinsic ambiguity such as homonyms and heteronyms [2]. Two laboratories that have the same name could research totally different topics, while two labs doing similar research could have completely different names. For example, in Korea, one lab with the name "HCI lab" belongs to the department of smart convergence technology and another lab with the same name belongs to the department of business. They are really different in terms of research topic. Information like the example above can often be misleading and make people unconfident about browsing results. That's why these two reasons necessarily lead to demanding decision making and deter or misdirect the seekers.

In the field of information theory, decision making is characterized by facing pre-defined problem and conceptualizing information needs which are not a precisely defined or structured [13]. Decision-making theories normally focus on how or why someone chooses a certain course of action and what determines that choice [3]. Based on the theory, many studies investigated empirical understanding of the decision-making process. Julien focused on similar difficulties

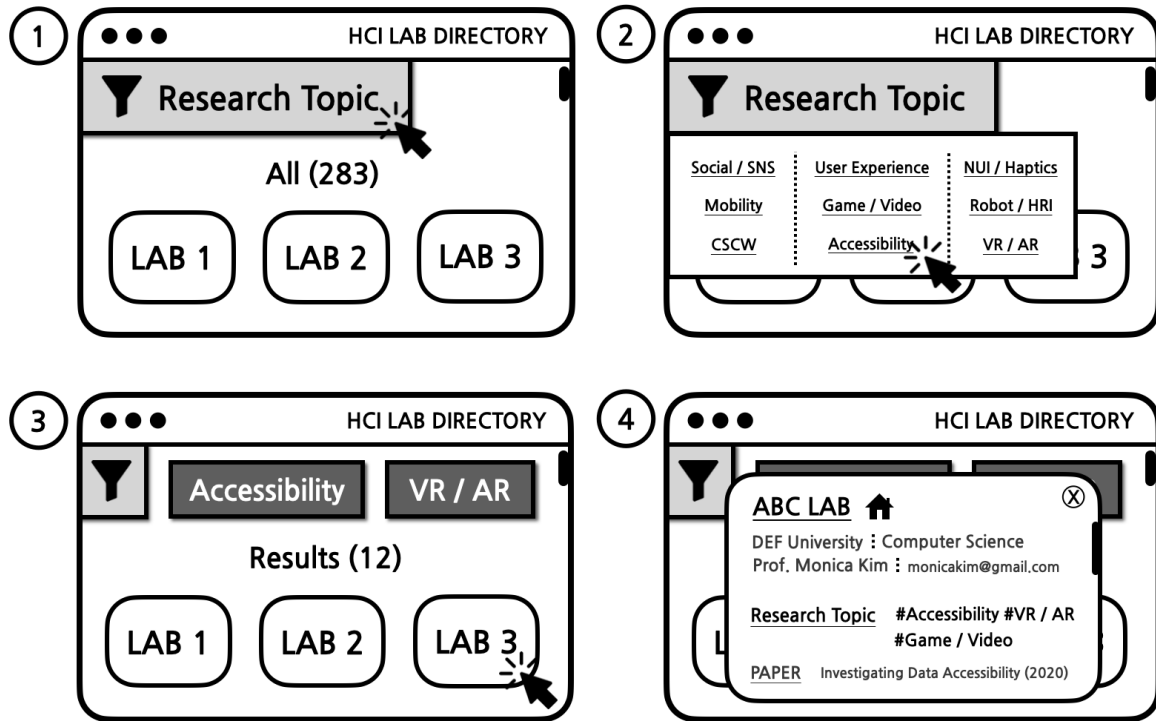


Figure 1: HCI LAB DIRECTORY : You can choose maximum five research topics together (2). After choosing topics, you can click the lab card on the results list (3). In the lab card pop-up window, you can check more information about publication and visit the website (4).

in decision making faced in terms of future career planning [5]. The results demonstrate that adolescents did not understand what decisions they needed to make about their futures since they had unclear notions of any information about career seeking. This lack of clarity led them to feel anxious and overwhelmed during the complex decision-making process. Also, previous works on web systems that support decision making studied the importance of information structure for decision quality and satisfaction, especially in information-rich environments [1, 9]. Moreover, Vlahos et al. analyzed that using computer-based information systems by German IT managers supported decision making at work in terms of resource allocation, evaluating alternatives, identifying problems, and making short-term decisions [12].

Previous studies emphasized that what to decide and what to seek is vague in the decision-making process and web system with well-organized information could support it. They analyzed the satisfaction and quality of decision in terms of information structure. However, our work differs from them by observing users' experience in our system and investigating how they resolve difficulties from the previous decision-making process.

With this background, our study aimed to (1) design an HCI lab browsing system named <HCI LAB DIRECTORY> to support demanding decision making and (2) investigate how the system helps decision making through a user study. Our findings demonstrate our system supported identifying which labs to shortlist at

the initial stage and developing some criteria for evaluating alternatives by learning general knowledge of the field at the drill-down stage. This study contributes to provide empirical understanding of decision-making theory and presents a new browsing system operating by ADK analysis that organize scattered and unstandardized laboratory information.

2 SYSTEM DESIGN : HCI LAB DIRECTORY

We designed an HCI lab browsing system website where users could browse information about 283 HCI laboratories in South Korea with 26 research topic categories that are newly defined as controlled keywords (Figure 1). For other HCI societies or conferences, labs are usually sorted by lab name or university in alphabetical order, not by research topic. Even if so, traditional classification would not reflect actual research topics of a lab that is actively changing. Accordingly, we argue that an HCI lab browsing system not only whose classification system is actively changing with automatically updated data, but also whose ordering is by research topic from published paper analysis, is necessary.

Previous work on categorization of keywords, sentences, and abstract text data extracted from papers applied appropriate methods to the characteristics of each dataset, such as semantic analysis, network analysis, qualitative coding, and statistical correlation analysis. Liu et al. mapped and visualized the intellectual landscape of the field of HCI through the use of co-word analysis [8]. Moro and Bielekova's study defined an indexing keyword as "the Navigational

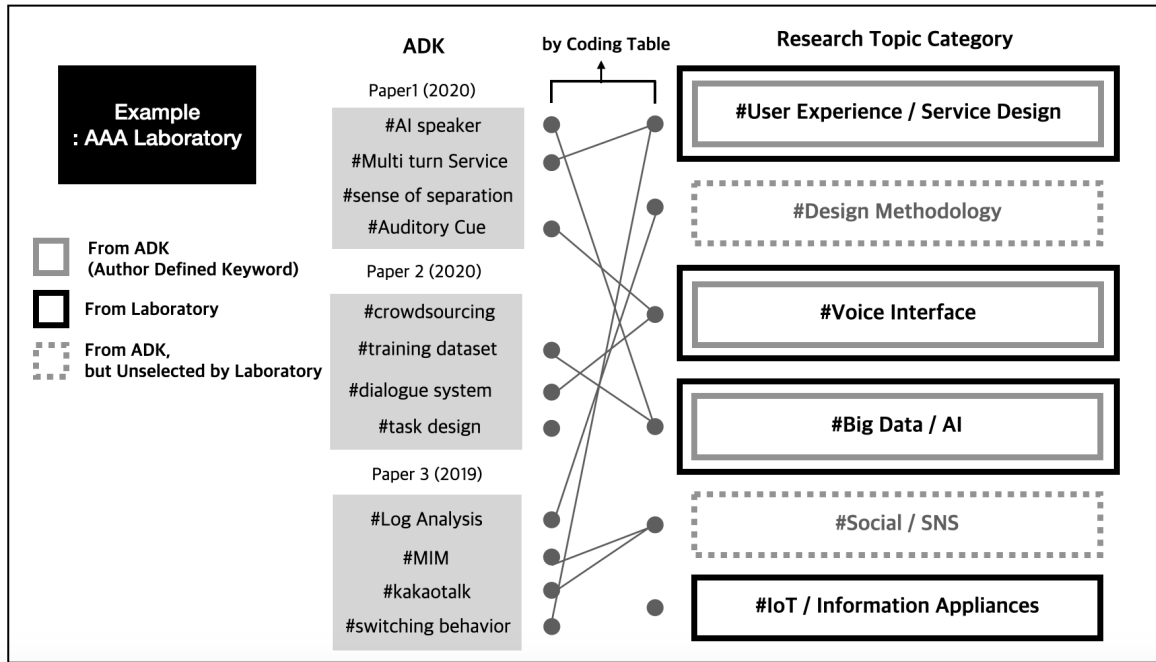


Figure 2: example of HCI laboratory mapping process under 26 research topic categories based on each lab's ADKs

Lead” that can cluster each type of thesis data into an LDA model and evaluated their performance [11]. Four types of data are keywords of abstract, keywords of the entire article, keywords defined by author, and keywords by readers.

However, publication data analysis in previous studies functions only as a category or indexing units and has not focused on information finding behavior led by the categorization. In this study, we attempted to define a set of indexing categories that is extracted from paper ADKs and to map the properties of each HCI laboratory under the same categories. Furthermore, we aimed to examine user experience of our new categories in the challenging decision-making process.

2.1 Defining Research Topic Category

We defined research topic categories that indicate properties of each laboratory. In order to involve the latest research trends, we modified paper review categories of HCI Korea (HCIK) based on grouping results of author keywords from recent 2 years HCIK articles. We decided to collect data from HCIK publication since HCI Korea is the most renowned academy in the field of HCI in Korea and has the highest number of articles that are published every year.

The 33 HCIK publication review categories which are classified into three areas of technology, design, and humanities have two limitations: First, dividing them into three areas is not suitable to cover the field of HCI studies, where interdisciplinary convergence studies are actively carried out. Second, it does not reflect changes in research trends over time. Consequently, this work used ADKs to break down the existing classification and reorganized them to 26 research topic categories that represent recent research trends.

2.2 Qualitative Coding of Author-Defined Keywords

We coded 5,299 ADKs, adapting 26 research topic categories defined in this study. Repeated keywords were coded automatically to the same category. Two researchers independently coded all the keywords, compared their codes, and discussed possible revisions. This process resulted in final inter-coder reliability of 82.6% [10]. After this process, we mapped 283 HCI labs into the maximum five categories according to the ADKs published by the lab. The data was limited to ADKs published in the HCIK conference and journal in the last five years (2016-2020). The reason why we focused on ADKs is that they are important content indicators that represent the author's intention rather than the other types of publication data [7].

However, ADKs have a limitation that its representativeness and quality depend on the level of experience and expertise of the author [7]. In order to compensate for this limitation, we asked professors, representatives from each lab for feedback about each lab's mapping result. About 30% of all labs requested modification of categories and three labs who requested the deletion were excluded. Figure 2 shows the example of research topic mapping process based on each lab's publication data.

3 METHOD

We conduct empirical investigation of decision-making process while browsing HCI lab information with our new system. We set a user study consisting of lab browsing task, pre-surveys, and semi-structured interviews.

3.1 Participants

Recruitment was conducted by convenience sampling. In order to recruit undergraduate students who are interested in entering the HCI graduate school and have past experience of browsing the lab on the web, we sent the announcement email using the email list of a lab to which the researchers belong. This was done to minimize the difference in usability that may vary for each research topic category by unifying the initial query when using this browsing system. We recruited 10 participants. (age $m = 25.7$, 7 women) Their undergraduate majors varied, including media communication, computer engineering, international commerce, graphic design, and business engineering.

3.2 Study Procedure

Our study had three steps which are (1) a lab browsing task using the system, (2) a survey before the task, and (3) semi-structured interview after the task. The entire study procedure took about 1.5 hour and carried out remotely through Zoom. Each participant received \$15 for participating in the study.

Before performing the given task, we asked 10 participants about past HCI lab browsing experience to understand each participant's level of interest and prior knowledge. After that, they conducted a task that is to find 3 interesting laboratories and their email address for more specific information. The time limit was minimum 15 minutes and maximum 45 minutes. They had to browse using our research topic categories, but other search engines were also available for further information. We asked them to use the think-aloud method while performing the task [6], so they could freely express their thoughts about the tasks in real time. The interview asked them about experience in browsing laboratories by research topic of interest (awareness of research topic categories, criteria for filtering information, task results and procedure) and past lab browsing experience (searching path, motivation, difficulty).

3.3 Analysis

We screen-recorded think-aloud and interview sessions. The qualitative data from the think-aloud sessions and interviews were transcribed and analyzed using grounded theory techniques [4].

4 FINDINGS

As a result of the user study, the success rate of the task which is finding three laboratories was 80%. Two participants out of ten failed because one participant has found just two labs and the other participant has exceeded the time limit. The average time to complete the task was about 28 minutes ($SD = 11\text{min}$). Participants clicked and looked through about 15 lab info cards on average ($SD = 7.4$).

We investigate what was previous lab browsing experience like and how it changed, eventually how our system helped participants' decision making. The difficulty of decision making and the implication carried by the system were different in the initial stage and the drill-down stage. Interview results demonstrate that our HCI lab browsing system supported participants' decision-making process in terms of (1) identifying and specifying what to look for at the initial stage and (2) developing one's own criteria by gaining saliency of the academic landscape at the drill-down stage.

4.1 Initial Stage : Identifying and Specifying What to Look For

According to the participants, the initial stage right after getting interested in HCI programs is characterized by "totally cluelessness", "pre-knowledge", and "university popularity". At first, they were just clueless about what to find and consider. They knew there are some HCI labs in Korea, but didn't know which keywords to search other than HCI lab. In addition, they were not sure if they could find the desired lab with the searching keywords they chose. P7 said it was difficult to start the process because she couldn't identify what to look for at all. However, due to the controlled categories presented by our browsing system, she was able to identify which one to browse from.

"I couldn't even start because I knew nothing. And even the same meaning can be expressed in multiple words, so I always had anxiety that I would not be able to find the lab I wanted due to the word I chose. But I think I have a sense of what to look at because of these standardized keywords." (P7)

Next, they cannot help depending on their pre-knowledge including their own experience, what they have heard from nearby acquaintances, and common sense. This led them to only see familiar majors or labs that they already know. However, using our new lab browsing system, P2, P6, and P10 identified the new research topic and dig into it where they had little interest before but did not take seriously due to lack of knowledge. They began to specify new alternatives and emerge from vague understanding of unfamiliar areas of the field:

"Back in the day, I was also interested in product design. But vaguely, I thought that I would have to major in a related department as an undergraduate... I didn't look it up. Actually, I didn't know much. After looking into several labs in this topic, I got interested enough to contact some." (P2)

Moreover, the state of cluelessness often made them browse from the university website or blog postings listed by popularity. Therefore, alternatives were often limited to a small number of universities. This is also because of frustration from previous search experiences. P8 said it is hard to figure out any lab in detail only with information such as the lab name or belonging department. What was worse, always only a few results were repeatedly presented. Thus, participants easily depended on the popularity of university at the initial stage. This could be the reason why most of them submitted new labs in the task results that belong to universities that they had never thought of before.

"Nowadays, there are so many labs with different names, so I don't know exactly what kind of research this lab is doing until I enter the website. So I guess I only checked a few famous unis and actually didn't even look at many labs." (P8)

4.2 Drill-Down Stage : Developing Own Criteria by Gaining Saliency of The Academic Landscape

At the drill-down stage, participants drill-down from general information such as lab name and department of university to detailed information such as paper, research projects, and other lab activities that indicate culture of the lab. At this stage, they face a state of chaos such that they cannot distinguish and compare numerous laboratories. P3 said that it was chaotic that he had to keep searching without defined target until discovering something interesting by simply continuing in and out of various laboratory webpages because he couldn't distinguish them:

"Even with the same UX, there are so many different labs of different topics and it's hard to distinguish what's different. It took a lot of time and was difficult because I had to just keep browsing one by one until something interesting came out" (P3)

The fact that basic information of the labs is not very helpful to distinguish and evaluate several alternatives makes it difficult to have confidence in the criteria developed at the drill-down stage. P8 explained that Lab AAA was the most interesting and satisfying discovery among the three labs submitted as a result of the task. This is because most of its recent research was related to her interest, mobility and autonomous driving. But she was quite certain that its main research topics could never have been inferred by the name of the lab alone or its superficial information.

"I skipped the AAA lab without thinking that it would be related to the mobility with the name. . . Then I just searched by this school and looked at what topics were there. Only then the mobility keyword of the AAA lab caught my eye, so I happened to see the lab." (P8)

However, our system supported participants learning and gaining saliency of academic fields, terms, research trends, topics, and methodologies. In their previous lab searching experiences, a similar learning effect may have been possible, but only by digging into specific lab websites yet understanding the overall trends of the field of HCI was not easy. This active learning process gave them clues for identifying what to consider and developing that criteria to evaluate labs. P9 didn't expect that there could be healthcare category. While browsing a lot of labs that are mapped with healthcare topic, he could learn research topics and trends about the area and applied that knowledge to the task results:

"I got another look at the research categories that I didn't even think about. When it's a healthcare area, it's very broad. I figured out this kind of healthcare research is the healthcare in HCI, and what they're studying these days. I submitted final 3 email address found by browsing labs that had 2 categories together, healthcare and UX." (P9)

In addition, the system links the lab with publication so brought participants to understand labs from the perspective of paper and research topic at the drill-down stage. The lab that they have had interest in is verified once again by newly developed criteria, and

it gives a sense of self-efficacy in decision making. P6 made sure of the suitability of the lab in which she's already been interested:

"The lab that I searched before published more papers than I thought. And then when I filtered by the categories I was interested in, it just came out. Checking that they matched, I got more interest for further search. It was very satisfying." (P6)

5 DISCUSSION AND CONCLUSION

This paper focuses on the demanding and inefficient decision-making experience for incoming graduate students in HCI who are looking for a laboratory. We designed a new browsing system and then conducted user study. According to the qualitative data analysis through user study, we investigated how it worked and supported participants' decision-making process. Our findings demonstrate our system supported identifying which labs to short-list at the initial stage and how to evaluate alternatives by learning saliency of the field at the drill-down stage. Our study provides empirical understanding regarding decision-making theory and presents a new browsing system that organizes scattered and un-standardized laboratory information by author-defined keywords from publication data.

However, it has the following limitation and needs improvement in future work. The sum of HCI Korea's publication data may not represent the research topic of the laboratory precisely. This is because the HCIK papers are just a part of the performance of the lab. About 30% of all labs requested adjustment of categorization, and this discrepancy shows that the data limited to the HCIK has a limitation in representing the research topic of each lab.

In future work, we will elaborate our browsing system by increasing publication data of the laboratories. In addition, we aim to investigate quantified decision-making process by analyzing lab browsing behavior data in our system for further understanding of the user experience.

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